Nomenclature

Dimensional variables

T	Temperature
Y_{O_2}, Y_H	Molar concentrations of O_2 and H radicals
x	Spatial coordinate
t	Time

Dimensional parameters

v	flame front speed
A_r, A_B	Rate constants of recombination and branching
D_{O_2}, D_H	Diffusion coefficients of oxidizer O_2 and H radicals
Q_r, Q_B	Specific heats released during recombination and branching
E	Activation energy of the chain-branching reaction
R	Gas constant
λ	Thermal conductivity
ρ	Density
c_p	Specific heat at constant pressure
Y_0	Oxidizer concentration in the fresh mixture
T_0	Initial (fresh mixture) temperature
T_b	Adiabatic flame temperature

Auxiliary dimensional parameters

$\alpha = \dfrac{\lambda}{\rho c_p}$	Thermal diffusivity
$u = \left(\dfrac{\alpha A_B Y_0}{\beta^2 e^\beta}\right)^{1/2}$	Characteristic velocity

Dimensionless parameters

$c = \dfrac{v}{u}$	Dimensionless flame front speed
$\beta = \dfrac{E}{RT_b}$	Dimensionless activation energy of the chain-branching reaction
$L_{O_2,H} = \dfrac{\alpha}{D_{A,B}}$	Lewis numbers for oxidizer O_2 and H radicals
$r = \dfrac{A_r \beta^2 e^\beta}{A_B}$	Dimensionless recombination rate
$\sigma = \dfrac{T_0}{T_b}$	Dimensionless initial temperature
$q = \dfrac{Q_r}{Q_r + Q_B}$	Fraction of heat released due to recombination

Dimensionless variables

$\theta = \dfrac{T}{T_b}$	Dimensionless temperature
$Y = \dfrac{Y_A}{Y_0}$	Dimensionless oxidizer concentration
$X = \dfrac{Y_B}{Y_0}$	Dimensionless radical concentration
$x' = x\dfrac{u}{\alpha}$	Dimensionless spatial coordinate
$t' = t\dfrac{u^2}{\alpha}$	Dimensionless time

The governing equations in dimensional form for the present model can be written as follows:

$$\begin{aligned}
\rho c_p \frac{\partial T}{\partial t} &= \lambda \frac{\partial^2 T}{\partial x^2} - v c_p \rho \frac{\partial T}{\partial x} + Q_r A_r Y_H Y_0 + Q_B A_B Y_{O_2} Y_H e^{-\frac{E}{RT}}, \\
\frac{\partial Y_{O_2}}{\partial t} &= D_{O_2} \frac{\partial^2 Y_{O_2}}{\partial x^2} - v \frac{\partial Y_{O_2}}{\partial x} - A_B Y_{O_2} Y_H e^{-\frac{E}{RT}}, \\
\frac{\partial Y_H}{\partial t} &= D_H \frac{\partial^2 Y_H}{\partial x^2} - v \frac{\partial Y_H}{\partial x} + A_B Y_{O_2} Y_H e^{-\frac{E}{RT}} - A_r Y_H Y_0,
\end{aligned} \tag{1}$$

The dimensional boundary conditions are

$$\begin{aligned}
T = T_0,\ Y_{O_2} = Y_0,\ Y_H = 0,\ x \to +\infty, \\
\partial T/\partial x = 0,\ \partial Y_{O_2}/\partial x = 0,\ Y_H = 0,\ x \to -\infty,
\end{aligned} \tag{2}$$

These conditions correspond to a combustion wave propagating in the positive direction of the x-axis, i.e., from left to right.

In nondimensional form, the equations take the following form:

$$\begin{aligned}
\frac{\partial \theta}{\partial t'} &= \frac{\partial^2 \theta}{\partial x'^2} - c \frac{\partial \theta}{\partial x'} + (1-\sigma)(qrX + (1-q)K_1), \\
\frac{\partial Y}{\partial t'} &= L_{O_2}^{-1} \frac{\partial^2 Y}{\partial x'^2} - c \frac{\partial Y}{\partial x'} - K_1, \\
\frac{\partial X}{\partial t'} &= L_H^{-1} \frac{\partial^2 X}{\partial x'^2} - c \frac{\partial X}{\partial x'} + K_1 - rX, \\
K_1 &= \beta^2 X Y e^{\beta(1-1/\theta)}.
\end{aligned} \tag{3}$$

The boundary conditions become

$$\begin{aligned}
\theta = \sigma,\ Y = 1,\ X = 0,\ x \to \infty, \\
\partial \theta/\partial x' = 0,\ \partial Y/\partial x' = 0,\ X = 0,\ x \to -\infty.
\end{aligned} \tag{4}$$

The principal parameters required for solving the nondimensional equations, their approximated dependence on the equivalence ratio, as well as the quantities necessary for transforming back to dimensional form, are listed in Table 1.

Table 1: Model parameters

Parameter	**Value**
E	70.3 kJ/mol
A_B	$2.2 \cdot 10^8$ m^3/(mol $\cdot$ s)
$T_b(\varphi)$	$-2.5860\varphi^3 + 58.4222\varphi^2 - 527.8977\varphi + 2899.3915$ K
$A_r(T_b/2)$	$6.06 \cdot 10^{12} \cdot (T_b/2)^{-3.42}$ m^3/(mol $\cdot$ s)
α	$1.9168 \cdot 10^{-7}\varphi^3 + 1.1869 \cdot 10^{-6}\varphi^2 - 8.2385 \cdot 10^{-5}\varphi + 8.4539 \cdot 10^{-4}$ m^2/s
Y_0	$\dfrac{p}{RT_0(4.77 + 2\varphi)}$ mol/m^3
$L_{O_2}(\varphi)$	$2.2979 \cdot 10^{-3}\varphi^3 - 4.3934 \cdot 10^{-2}\varphi^2 + 2.7184 \cdot 10^{-1}\varphi + 1.6759$
$L_H(\varphi)$	$3.8905 \cdot 10^{-4}\varphi^3 - 8.2983 \cdot 10^{-3}\varphi^2 + 6.708 \cdot 10^{-2}\varphi + 2.2724 \cdot 10^{-1}$
$r(\beta)$	$1.1 \cdot 10^{-8} \cdot \beta^{5.42} \cdot e^{\beta}$
q	0.9

Under the activation energy and pre-exponential factor in the Arrhenius law for the branching reaction, we refer to the corresponding parameters for the reaction $O_2 + H \to OH + O$. The values of E and A_B are taken from [1] (see Table 1).

The adiabatic flame temperature T_b, determined by the equivalence ratio φ, can be evaluated using the numerical package Cantera. The dependence $T_b(\varphi)$ can be accurately approximated by a third-degree polynomial in φ (see Table 1).

In the present model, the quantities ρ, c_p, and λ are assumed to be constant along the flame profile. Their values are taken at the temperature $T_b/2$ and pressure 1 atm from the NIST database [2]. The thermal diffusivity of the mixture α is calculated as a weighted average of the thermal diffusivities of the components N_2, H_2, and O_2, with weights equal to their respective mole fractions in the fresh mixture (see Table 1).

The dependence of the Lewis numbers for the oxidizer O_2 and the radicals H on φ, as well as the assumption that the parameter q is constant and equal to 0.9, are taken from [3]. The dependence of the Lewis numbers on the equivalence ratio is approximated by a cubic polynomial in φ (see Table 1).

The recombination rate r is also assumed to be spatially constant, with its value evaluated at the temperature $T_b/2$. In the nondimensional formulation, it is convenient to use the dependence $r(\beta)$, whereas in dimensional form the expression $A_r(T_b/2)$ is employed. This dependence was fitted to best reproduce the experimental data for the flame speed as a function of the equivalence ratio (Fig. 1) and the radical concentration profiles in comparison with GRI3.0 (Fig. 2). It should be noted that the dependence $r(\beta)$ was obtained for a fresh mixture temperature $T_0 = 298$ K and pressure $p = 1$ atm.

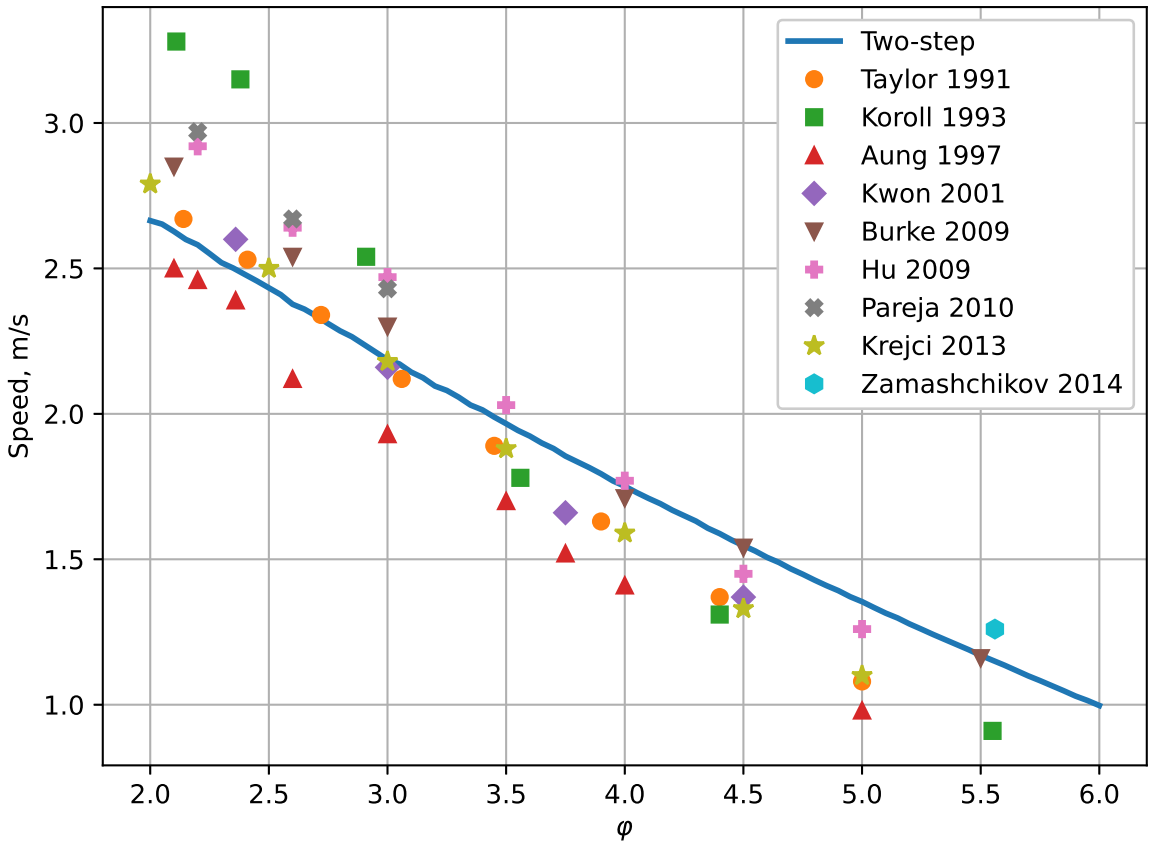


Figure 1: Dependence of the flame front velocity on the equivalence ratio at a fresh mixture temperature of 298 K and a pressure of 1 atm. The solid line represents the results of the two-step model, while the colored markers denote the experimental data [4–12] from the review [13].

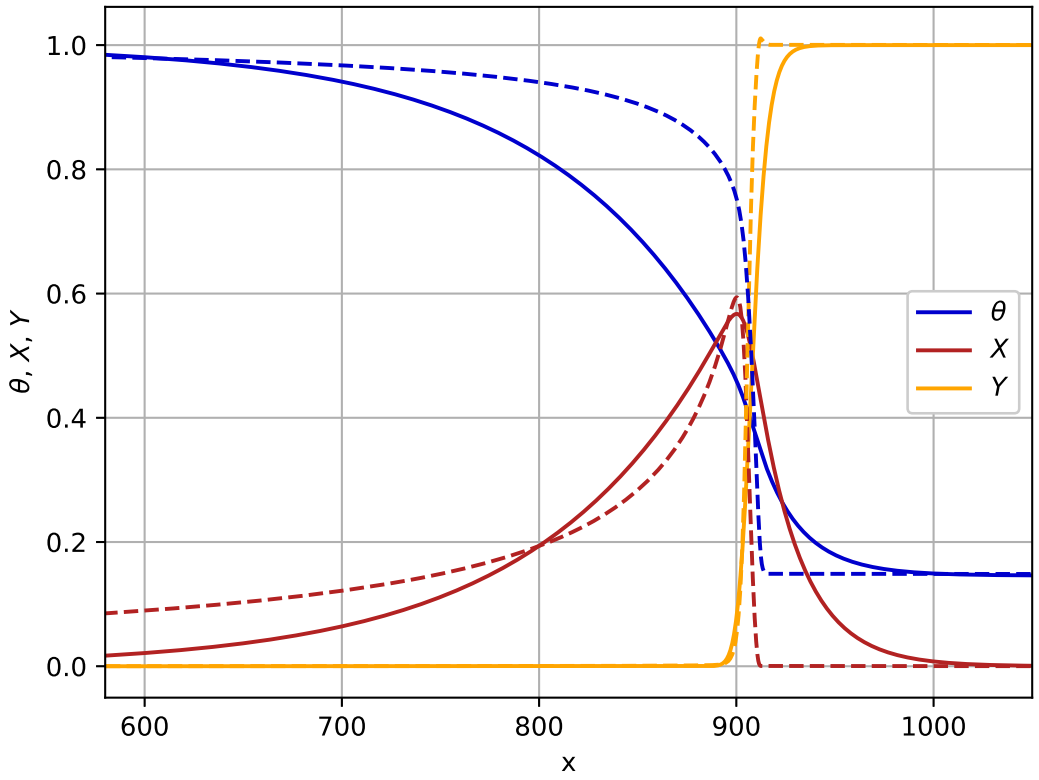


Figure 2: Nondimensional temperature and concentration profiles at a fresh mixture temperature of 298 K, a pressure of 1 atm, and an equivalence ratio of $\varphi = 2.1$. The solid lines represent the results of the two-step model, while the dashed lines correspond to calculations performed using GRI 3.0.

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